

The Ubiquity of Tiebout Sorting Across Neighborhood Boundaries

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Abstract

Neighborhood boundaries are often determined by physical topography, transportation networks, or the administration of public goods. We present a simple model of boundaries that predicts spatial discontinuities in household demographics, the supply of amenities, and home prices. We take these predictions to the data and find abundant evidence of discontinuities in a wide range of observable dimensions and six different types of boundaries. We draw two conclusions: (1) researchers should implement boundary discontinuity designs with caution because the key identification assumption may not hold except in specific applications, and (2) even narrowly targeted place-based policies may have broader dynamic impacts.

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1 Introduction

Cities are subdivided into neighborhoods by boundaries that can be natural, physical, political and administrative. In principle, boundaries may partition cities into noticeably distinct neighborhoods characterized by the interactions of households, firms, and local governments. Alternatively, boundaries may simply exist in parallel to the lived experiences of local residents and go otherwise unnoticed. In the former view, a rich literature following Tiebout (1956) has studied both the process of¹ and the consequences of² neighborhood sorting. Fundamental to this literature is the idea that neighborhood boundaries represent more than simply lines on a map, they induce distortions that impact decision makers, making it difficult to attribute differences across boundaries to a single source. Meanwhile, a relatively young literature in empirical social science has embraced the alternative view in hopes of leveraging boundaries for causal inference.³ Ultimately, it is an empirical question as to which of the two views best describes reality, the answer of which contributes to our fundamental understanding of the distribution of economic activity across space (Clark et al., 2003).

The extent to which boundaries propagate Tiebout sorting highlights a tension between behavioral responses of economic agents and policy evaluation using boundaries. In spatial econometrics, boundaries represent geographic thresholds where differences in outcomes are (plausibly) only caused by a treatment of interest. So long as unobserved confounders vary smoothly across the boundary space, these models produce estimates that meet a high bar for internal validity (Keele and Titunik, 2015; Butts, 2023). The challenge for researchers is the long-held intuition that adjacent neighborhoods separated by boundaries differ in a variety of ways,⁴ and only rough proxies for neighborhood demographics are typically available to rule out Tiebout sorting. Notwithstanding, the rich tapestry of boundaries in the urban landscape lends to research evaluating access to public goods and services, exposure to particular place-based interventions, proximity to private

¹See, for example, Epple et al. (1984, 2003); Bayer et al. (2004); Caetano (2019); Caetano and Maheshri (2023)

²See, for example, Black (1999); Bayer et al. (2007); Chetty et al. (2018)

³The strategy of comparing groups on opposite sides of a boundary to identify the effects of an intervention underlies a broad class of empirical approaches that includes boundary fixed effects, spatial difference-in-differences, and boundary discontinuity designs.

⁴The notion of a “wrong side of the railroad tracks” is at least a century old, having found its way into a wedding announcement in the *Daily Inter Ocean* (Chicago) in 1903.

consumption opportunities, residential segregation, and network effects.

It is perhaps surprising then that comprehensive empirical evidence of Tiebout sorting (or the lack thereof) at neighborhood boundaries does not exist despite the broad appeal of spatial models exploiting boundaries for identification. This could be because the estimation of boundary models is data intensive, and the comparative advantage of boundary designs - sharp inference about a particular policy affecting a small area of interest - offers little incentive to conduct comprehensive analysis. This second reason falls under a more general critique of the external validity of boundary designs, as researchers must only rule out sorting and other potential confounders unique to the setting of interest, trading off the license to make broader claims.

In this paper we present a simple theoretical framework and a wealth of evidence to understand how boundaries cause Tiebout sorting and its empirical implications. Our approach begins with a stylized model that illustrates how boundaries distort the location decisions of households, which in turn distort the amenities supplied by private firms and the public goods provided by local governments. The key empirical insight of the model is that these distortions will manifest as discontinuities in the demographic and amenity bundle of neighborhoods and ultimately in neighborhood prices. We then test for the existence of spatial discontinuities in the universe of all publicly available variables from the US Census, analyzing the demographics of block groups near six very different types of boundaries: historical rail and highway networks, contemporary school district boundaries and attendance zones, county lines, and ZIP code boundaries. Consistent with predictions of the model, we find overwhelming evidence of discontinuities across a wide range of variables and all types of boundaries.

Our model is a simple formalization of the insights of two seminal papers on sorting. Following Hotelling (1929), we start from the notion that certain consumers wish to locate close to certain producers in horizontally differentiated markets. We then incorporate Tiebout (1956) sorting stemming from heterogeneity in public goods supplied across neighborhoods. The basic conclusion of the model is that households of a given type will cluster on the same side of a boundary that provides proximity to their preferred private and public goods. A core prediction is that this clustering will generate a discontinuity at the boundary; the endogenous responses of households,

firms, and local governments create a feedback loop that amplify this discontinuity and hence the causal effects of boundaries on local demographics and amenities. We show that the magnitude of this problem for boundary designs depend on the timing of boundary placement (or treatment assignment to each side) relative to the timing of the explanatory variable of interest. This implies that boundary designs are not irredeemable for empirical analysis, but rather they are more appropriate to identify the effects of variables that change (and are observed) immediately after the placement (or deletion) of a new boundary.

Following our model, the econometric framework to test for Tiebout sorting at boundaries simply involves the estimation of a boundary discontinuity design.. We implement this test with a dataset comprised of detailed maps of the six aforementioned boundaries throughout the contiguous US and all variables in all census block groups in the US in 2010 and 2020. The average block group contains 1395 residents, and there are over 1700 census variables that describe their demographic characteristics along with features of the housing stock, labor market profiles, and measures of government assistance take-up. Each block group is assigned to a nearby boundary based on distance, and each census variable is taken as an outcome of the model, separately for each boundary type. All types of neighborhood characteristics systematically vary discontinuously at all types of boundaries, with the largest discontinuities found at school zones, where 81% of characteristics vary discontinuously, and railroad tracks, where 76% of characteristics vary discontinuously. We also find robust evidence of Tiebout sorting at ZIP code (64%) and school district (61%) boundaries, and weaker evidence at highway (51%) and county lines (17%).

Standard approaches to estimate boundary effects require researchers to know which side of a boundary is treated and which side of a boundary is untreated. These approaches then can then identify treatment effects in a regression discontinuity framework where the running variable is distance to boundary (untreated neighborhoods are usually assigned negative distances, and treated neighborhoods are usually assigned positive distances). In our setting, we do not have *a priori* treated and untreated sides of boundaries; our goal is simply to identify discontinuities and estimate their magnitudes. Moreover, we seek to estimate these discontinuities on vast, highly intersecting boundary networks that span the entire United States.

For these reasons, we must modify the standard approach to our needs. We do so by proposing a fully data driven procedure to spatially orient all segments of all boundaries in a consistent manner for estimation. Briefly, we identify the “high side” of a boundary for a given characteristic as the side where neighborhoods tend to feature higher levels of that characteristic, and then we orient all neighborhoods in their distance to the boundary from the farthest neighborhood on the “low side” to the farthest neighborhood on the “high side.” We provide several pieces of evidence that establish that discontinuities will not be spuriously identified with such a procedure. Specifically, we conduct a series of placebo tests, conduct a Monte Carlo exercise, and provide direct empirical evidence that our approach is largely insensitive to changes in the orientation criterion that selects the high sides of boundaries.

These findings have important implications for the broad class of empirical strategies that exploit geographic boundaries for identification. We show that the credibility of boundary-based research designs depends critically on the timing of treatment relative to the sorting process. When a new boundary is introduced and the treatment is observed before households, firms, and local governments have time to adjust, boundary discontinuity designs can credibly isolate the effects of a single neighborhood characteristic. However, as the length of time between boundary placement and the period in which the variable of interest is observed increases, researchers will have to impose stronger and stronger identifying assumptions that are increasingly at odds with the empirical evidence that we present.⁵ Moreover, when new policies are assigned using long-standing administrative boundaries, the identifying variation reflects changes in the bundle of neighborhood characteristics, rather than a single treatment.⁶

Our framework shifts the interpretation of hedonic models using neighborhood boundaries. Rather than willingness to pay for a single characteristic, housing values capitalize conditions that

⁵The literature on persistence provides careful examination of contemporary economic outcomes at historic boundaries determining exposure to past institutions (Dell, 2010; Dell et al., 2018; Bühler, 2023). Additionally, property rights assigned by political lines have long-run effects on agricultural production and land use (Baragwanath and Bayi, 2020; Bühler, 2023).

⁶The literature on minimum wages has adopted the idea that labor markets on each side of boundary (ie state or county line) must adjust when one jurisdiction adopts new policies (Dube et al., 2010; Dharmasankar and Yoo, 2023). Jardim et al. (2024) argue that geographically diffuse spillovers of a minimum wage enacted on one side of a boundary affect wages and working hours on the opposite side, potentially contaminating the estimated employment impacts. A second example are place-based policies to stimulate economic growth, with explicit goals to induce positive income sorting and firm location choices (Lu et al., 2019; Albertus, 2020; Jia et al., 2021a).

emerge from household sorting and adjustments by firms and local governments. Using nationally representative school attendance boundaries, where we find the largest Tiebout effects, we estimate a house price premium of 25–33 percent to live on the high-amenity side. This full on board price reflects differentials in school characteristics, neighborhood composition, private consumption, and a host of unobserved amenities affecting estimates for the capitalization of school quality alone.

The remainder of this paper is organized as follows. In Section 2, we present a simple model of consumer location decisions, and we characterize how they are affected by physical and administrative boundaries. In Section 3, we present our empirical approach to estimate discontinuities in a large set of variables with no *a priori* spatial ordering, and describe the various sources of data that we use to implement this approach. In Section 4 we present our results and the implications for practitioners using boundary designs. We conclude in Section 5.

2 Conceptual Framework

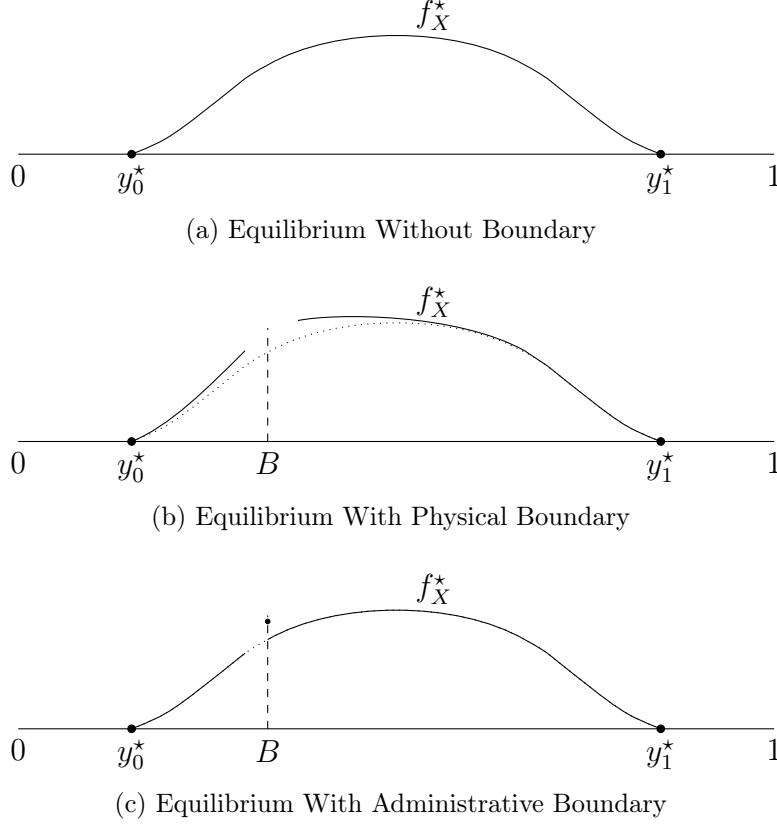
We motivate how boundaries generate discontinuities in households’ location choices in a simple, illustrative model. Boundaries are modeled as either physical distortions that increase the distance between points on opposite sides of the boundary, or mechanisms that allow for different levels of public goods to be supplied (or both). For simplicity, we assume that these boundaries are exogenously located along with two producers who are exogenously located at points $0 < y_0 \leq y_1 < 1$ on the unit interval, and a unit mass of consumers, indexed by i , who endogenously locate at points $x_i \in [0, 1]$. There is also a public good that is supplied exogenously at a level of g .

Let $d_i^j = |x_i - y_j|$ be the distance between consumer i and producer j . Then consumer i ’s utility is given by

$$U(x_i) = \alpha_i u(d_i^0) + (1 - \alpha_i) u(d_i^1) + v(g) \quad (1)$$

where the parameter α_i is drawn from a single peaked distribution over $(0, 1)$. We assume that $u' < 0$ and $u'' > 0$. That is, α_i represents consumer i ’s relative preference for producer 0 to producer 1, and all consumers prefer locating closer to producers (with a diminishing loss in marginal utility in distance to producers). We also assume that $v' > 0$.

Figure 1: Spatial Equilibrium



Notes: Panel A illustrates the distribution of optimal consumer location choices f_X^* , given heterogeneous preferences over distance to producers y_0^* and y_1^* . Panel B and C illustrate predictions from a model of consumer behavior when a boundary occurs at point B, relative to the initial distribution shown by the dotted line.

Consumer i chooses x_i to maximize the objective in equation (1). The first order condition implies

$$\frac{\alpha_i}{(1 - \alpha_i)} \frac{u'(d_i^0)}{u'(d_i^1)} = 1 \quad (2)$$

in equilibrium. This has a familiar interpretation, as the left-hand side of equation (2) is the marginal rate of substitution between the two producers. The right-hand side of the equation corresponds to the price ratio if we understand distances to producers to be effective prices since $-\frac{\partial d_i^1}{\partial x_i} / \frac{\partial d_i^0}{\partial x_i} = 1$. No consumer will ever locate outside of the interval $[y_0, y_1]$ as they would be strictly better off moving into the interval. Hence, we can illustrate the spatial equilibrium in the top panel of Figure 1.

We now introduce an exogenous boundary at some point $B \in (y_0, y_1)$. A basic characteristic

of many boundaries such as highways or rivers is that they distort the physical environment. We capture this by modeling the distance between any two points on opposite sides of B increased by $\beta \geq 0$. Consumer i 's first order condition is now

$$\frac{\alpha_i}{(1 - \alpha_i)} \frac{u'(d_i^0)}{u'(d_i^1 + \beta)} = 1 \quad x_i < B \quad (3)$$

$$\frac{\alpha_i}{(1 - \alpha_i)} \frac{u'(d_i^0 + \beta)}{u'(d_i^1)} = 1 \quad x_i > B \quad (4)$$

All consumers with $x_i^* < B$ before the introduction of the boundary remain to the left of the boundary and vice versa. The distortion has the effect of shifting the mass of consumers away from the boundary with a greater shift for those consumers who are located farthest from the boundary. We illustrate this in panel (b) of Figure 1. In general, physical boundaries generate discontinuities in the locations of consumers.

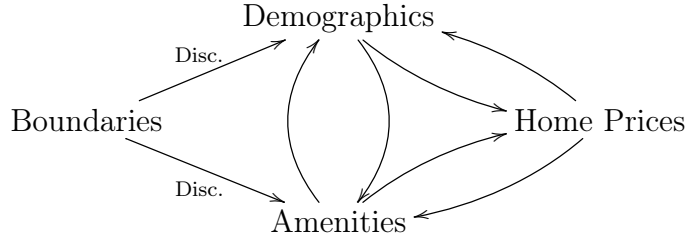
A second characteristic of many boundaries such as school zones or political borders is that they allow for public goods to be differentiated.⁷ We model this by specifying $g(x) = g_0$ at all points $x \leq B$ and $g(x) = g_1$ at all points $x > B$ where $g_0 < g_1$ without loss of generality. This has the effect of shifting a mass of consumers just to the left of B across the boundary. We illustrate the effects of an administrative boundary on spatial equilibrium in panel (c) of Figure 1. In general, administrative boundaries also generate discontinuities in the locations of consumers. Of course, many boundaries are both physical and administrative. For example, political boundaries may coincide with rivers or school district boundaries may coincide with roadways. This does not affect the qualitative conclusions of our analysis. Moreover, these conclusions will persist and likely strengthen if we endogenized the locations of boundaries or producers or if we endogenized the levels of public goods since this would increase incentives for sorting.

If consumers belonged to different demographic groups and these groups had systematically different tastes for the producers (the distributions of α_i differed across groups) or different tastes for public goods ($v(\cdot)$ differed across groups) then this simple analysis would yield further insights. Because boundaries would generate discontinuities in the locations of both groups of consumers,

⁷To simplify notation, we assume consumers exactly at the boundary can choose the side of the boundary to which they locate.

then we would generically observe discontinuities in demographic compositions across boundaries. Moreover, if producers adjusted their outputs to cater to their clientele, then this would imply discontinuities in the amenities supplied across boundaries. Similarly, to the extent that governments respond to the preferences of their constituencies, then this would imply discontinuities in the public goods supplied across boundaries. Finally, as amenities and public goods are capitalized into home prices, this may affect the demographics of new consumers.

We summarize how each of these mechanisms contributes to the positive feedback loops shown in the following diagram:



Boundaries generate discontinuities in demographics and amenities. Demographics and amenities are then co-determined with home prices through the sorting of households (consumers) and adjustments made by the suppliers of private amenities (producers) and public amenities (governments). Even if this sorting process is entirely continuous, the ultimate effect of a boundary on the urban landscape will be discontinuous in the characteristics of residents, characteristics of the amenity bundle, and prices.

3 Empirical Approach

3.1 Estimating Boundary Discontinuities Without Treatment

We observe a boundary network (e.g., the interstate highway network) as a series of curves in space, and we observe a vector C of local characteristics (e.g., population demographics, house prices) at a set of discrete points in space, which, in an abuse of nomenclature, we refer to as neighborhoods. We index these neighborhoods with j . For a given boundary network, we divide all boundaries into small, equal length pieces, which we denote as boundary segments indexed by

b. For each boundary segment, we consider a set of nearby neighborhoods denoted as J_b .

Let c be an element of the local characteristic vector C (e.g., house price). Our goal is to identify whether c varies discontinuously at boundaries. In a standard boundary discontinuity design (BDD), under certain assumptions, this potential discontinuity would correspond to the causal effect of a treatment that was applied to neighborhoods on a known side of each boundary segment. A researcher would estimate how c varies in distance to the boundary separately from the treated side and the untreated side, and in the limit at the boundary, the difference in this estimate of c from each side, i.e., the size of the discontinuity at the boundary, yields the treatment effect.

In contrast, we ascribe no causal interpretation to such a discontinuity precisely because no specific treatment has been applied at this boundary. Although we observe the spatial relationship of each neighborhood in each J_b to the relevant boundary segment b as in a BDD, we cannot use that approach because we have no information on the relationships between neighborhoods in different neighborhood sets J_b . Specifically, in a BDD, two neighborhoods on the treated sides of the boundaries in J_b and $J_{b'}$ both share a common characteristic of being treated. Meanwhile, in our context, two neighborhoods on some side of boundaries in J_b and $J_{b'}$ do not share any characteristics in common *a priori*. If we were to use some arbitrary approach to assign “treatment” status, e.g., the South sides of boundaries are treated, then we would fail to identify a discontinuity in c if the gradient of c was increasing in a Southerly direction for half of the boundary segments and was decreasing in a Southerly direction for the other half.

This example indicates that we need a purely empirically driven approach to ordering the direction of boundaries. We do so by locating all neighborhoods in J_b within a distance of λ on either side of the boundary and refer to them as index neighborhoods. For each neighborhood characteristic c , boundary segment b , and neighborhood $j \in J_b$, we construct a dummy $H_{cbj} = 1$ for all neighborhoods that are on the same side of the boundary as the index neighborhood with a higher value of c (i.e., the “high side”) and $H_{cbj} = 0$ otherwise. For each $j \in J_b$ on the high side, we define d_{cbj} to be the distance from j to the boundary, and for each j on the low side, we define d_{cbj}

to be -1 times the distance from j to the boundary. We then estimate the following regression:

$$c_{bj} = \delta_c H_{cbj} + f_-(d_{cbj}) \times 1(d_{cbj} < 0) + f_+(d_{cbj}) \times 1(d_{cbj} > 0) + \epsilon_{cbj} \quad (5)$$

where $f_-(\cdot)$ and $f_+(\cdot)$ are flexible functions of the distance to the boundary and ϵ_{cbj} is an error term. The parameter δ_c corresponds to the boundary discontinuity for characteristic c .

This approach to orienting the direction of boundaries mechanically tends to assign higher values of c_{bj} to positive values of d_{cbj} and lower values of c_{bj} to negative values of d_{cbj} . However, it does not induce a *discontinuity* in c_{bj} at the boundary. To establish this, we perform three separate validation exercises of our method, described below:

Placebo Validation Exercise

The most intuitive empirical test of the ability of our empirical approach to avoid finding discontinuities when none should exist involves a placebo exercise in which fake boundaries are drawn. To simulate placebo boundaries, we shift every neighborhood in a boundary segment by 0.5 miles to the left or right in the two dimensional boundary space with the direction of the shift determined randomly. That is, we reshuffle the position of each neighborhood $j \in J_b$ by $\xi_{cb} = \{-0.5, 0.5\}$ with equal probability. For each characteristic c we estimate the placebo regression

$$c_{bj} = \delta_c^p H_{cbj} + f_-^p(d_{cbj} + \xi_{cb}) \times 1(d_{cbj} + \xi_{cb} < 0) + f_+^p(d_{cbj} + \xi_{cb}) \times 1(d_{cbj} + \xi_{cb} > 0) + \epsilon_{cbj}^p \quad (6)$$

which can be understood as an analog to equation (5) in which the location of each boundary segment has been randomly assigned across the map containing neighborhoods $j \in J_b$. If our approach is sound, then we would expect our estimate of δ_c^p to be zero.

Sensitivity of the Boundary Orientation Selection Procedure

The “high side” selection procedure described above involves using single index neighborhoods to determine the orientation of boundaries. We modify this procedure by using the entire set of neighborhoods within a given distance d of a boundary as index neighborhoods. Specifically, we

locate all neighborhoods in J_b on either side of the boundary that lie within d of the boundary. For each neighborhood characteristic c , boundary segment b , and neighborhood $j \in J_b$, we average the value of c across all neighborhoods within that set on either side of the boundary. We then define the dummy $H_{cbj} = 1$ for all neighborhoods that are on the same side of the boundary with the higher value of this average (i.e., the “high side”) and $H_{cbj} = 0$ otherwise. We repeat this test for various values of d . If our approach is sound, then we would expect our estimates of boundary discontinuities to be insensitive to our choice of d .

Monte Carlo

We can conduct a series of Monte Carlo validation exercises tests to determine whether our empirical approach erroneously finds discontinuities when none exist and to assess the extent that it identifies discontinuities when they do exist. Briefly, we generate hypothetical data with a known spatial gradient and across boundaries with random orientations where some of these simulated data have boundary discontinuities, and others do not. Full details and results of the Monte Carlo is available in Appendix B.1.

Bandwidth and Local Polynomial Selection

Finally, we examine the sensitivity of our empirical procedure to the choice of bandwidth and local polynomial specification. In our main results we employ the bandwidth selection technique proposed by Imbens and Kalyanaraman (2012). Because this approach optimizes each regression based on data availability of the outcome variable, we re-estimate all boundary discontinuities using various fixed bandwidths as common in the literature. Additionally, we report baseline results using a third-order polynomial to account for nonlinear spatial gradients in the different outcome variables, and show that our results are robust to this specification choice. We present these sensitivity analyses in Appendix B.2.

3.2 Data

To estimate the boundary discontinuities δ_C , we construct a dataset that is comprised of census block groups geospatially merged to nearby boundaries. Latitude and longitude estimates for the center of population of each block group is provided by the National Historical Geographic Information System (NHGIS), and we use ARCGIS software to map population centers to 2 mile segments of each boundary type. The rich set of publicly available data at the census block group level allows us to describe the area near each boundary segment along more than one thousand dimensions. We describe both the boundary network data and neighborhood variables in further detail.

3.2.1 Boundary Networks

We analyze three types of boundaries: transportation networks, educational boundaries, and administrative boundaries. Transportation networks form physical boundaries as they deform the urban landscape, are costly to cross, and often delineate distinct neighborhoods. We consider historical rail and highway networks, which serve as examples of both shrinking and growing transportation networks respectively. The US railway network peaked at 254,000 miles of track in the early twentieth century, and today it is comprised of approximately 160,500 miles of track.⁸ We measure the historical US rail network using the Atack (2013) historic GIS transportation database created from the New Century Atlas maps published in 1911. This includes all passenger and freight rail lines that were in operation circa 2011. The Interstate Highway System stretches nearly 50,000 and is part of a larger network that includes state highways. For the analysis we employ a digitized map of only interstate highways made publicly available by the US Department of Transportation as of 2020.⁹

School district and school attendance zone boundaries are constructed using shapefiles provided by NHGIS. The National Center for Education Statistics (NCES) conducts an annual update of school district boundaries dating back to 1995, and we obtain boundaries from the 2020 update.

⁸American Association of Railroads, *Chronology of America's Freight Railroads*. <https://www.aar.org/chronology-of-americas-freight-railroads/>.

⁹Our analysis omits segments of interstate < 0.5 mile long. Additionally, state managed highways are not present in the shape file.

Unlike school districts, there is scant nationally representative spatial data for school attendance zones. NHGIS hosts the 2009-2010 shapefiles created by the NSF funded SABINS project. The analysis of school attendance boundaries is restricted to elementary schools, which are defined as those enrolling third grade students (and none above ninth grade). Because the best available data for US school boundaries are from the 2009-2010 school year, we estimate this model with census block group data from 2010.

We also consider administrative boundaries in the form of 2020 county lines and ZIP code boundaries. A large share of county boundaries exist in remote areas with few nearby block groups, however, our sample is restricted to boundary segments with descriptive census data on both sides. Such a restriction results in smaller samples for county line regressions, despite the near 100% geographic coverage. ZIP code shapefiles also have extensive geographic coverage, but they contrast with county lines in that the data restriction is far less binding.

3.2.2 Sample Description

We provide summary statistics of our data in Table 1, where we compare selected neighborhood characteristics in each boundary network. All block groups are within one mile of each boundary type, with the sample restricted to boundaries with at least two block groups on either side. For five of the six boundaries we use block group data from the 2020 Decennial Census and the 2016-2020 American Community Survey (ACS). For school zones we use 2010 Decennial Census block group data. For 2020, there are over 1,700 variables characterizing the people, housing, and amenities for each block group in the sample contiguous 48 states (with over 2,000 for 2010). Tens of millions of people live within a mile of our sample boundaries with the most being near railroads (98.7 million) and the least living near county lines (25.9 million). On average, block group population means range from 1,674 to 1,859 compared to a population average of 1,776. The variation in home values and income across boundary types reflect the exclusion of boundaries in rural areas that do not meet the sampling restriction. Incomes for the sample (\$77,768) are relatively higher than US national averages (\$67,521) at the time.

Table 1: Block Group Summary Statistics

Variables	Full Data	Railroad	Highway	School District	School Zone	County Line	ZIP Code
Home Value	304,807.22 (249299.8)	298,811.16 (257542.2)	340,305.52 (275536.2)	329,255.35 (263742.7)	307,648.80 (211625.8)	340,140.41 (266735.6)	331,725.58 (271314.8)
Income	77,768.43 (40300.6)	74,642.87 (39464.3)	79,372.11 (40816.8)	85,446.41 (42854.5)	74,813.60 (38145.6)	87,853.63 (43871.7)	81,567.33 (41879.6)
Share White	0.60 (0.302)	0.59 (0.305)	0.53 (0.305)	0.61 (0.300)	0.62 (0.298)	0.63 (0.291)	0.58 (0.304)
Share Latino	0.19 (0.235)	0.19 (0.239)	0.21 (0.245)	0.17 (0.227)	0.18 (0.227)	0.14 (0.180)	0.19 (0.237)
Share Black	0.12 (0.201)	0.12 (0.207)	0.14 (0.217)	0.11 (0.194)	0.13 (0.213)	0.13 (0.219)	0.12 (0.201)
Rooms / Unit	5.82 (1.374)	5.65 (1.316)	5.61 (1.399)	5.97 (1.471)	5.78 (1.272)	6.01 (1.562)	5.85 (1.447)
Age of Housing	52.96 (159.1)	55.73 (143.3)	56.00 (152.9)	60.05 (184.4)	35.33 (17.98)	60.69 (190.6)	59.06 (181.3)
Observations	65639	56815	39875	33685	35604	14266	45765

Notes: Means and standard deviations (in parentheses) for the full sample of census block groups that lie within 1 mile of any boundary (column (2)), and the six subsamples of block groups for each corresponding boundary network.) Because school zone boundaries reflect the 2009-2010 school year, census data for this column come from the 2010 Census. For all other boundaries the data are from the 2020 Census. Income and home value measures displayed are in 2020 dollars.

4 Results

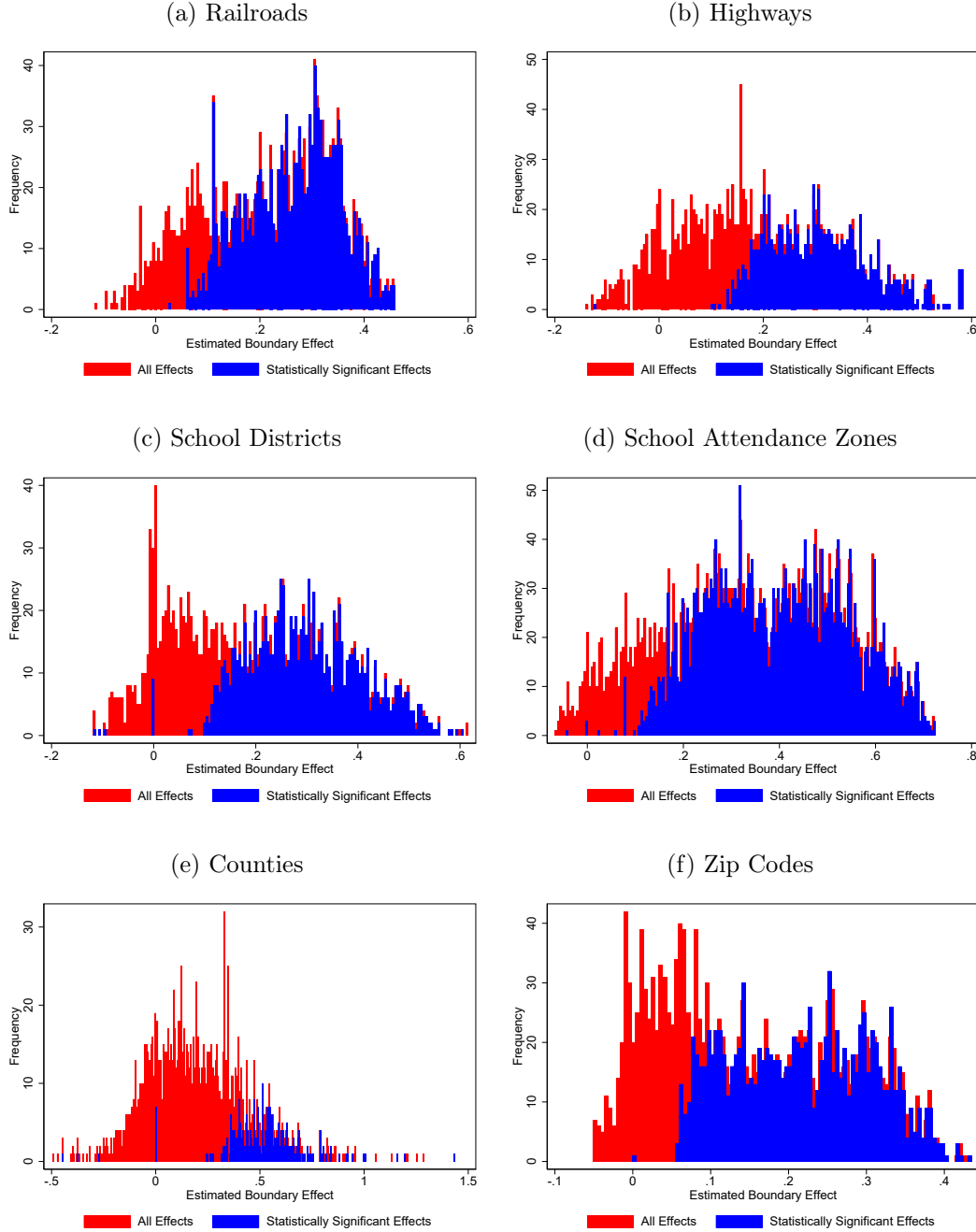
4.1 Tiebout Sorting By Boundary Type

Our primary empirical exercise in this paper is to estimate the share of census variables that exhibit statistically significant discontinuities at boundaries. For each boundary type, we present histograms of all estimated boundary discontinuities in Figure 2 overlaid with histograms of those effects that are significant at the 95% level. In order to facilitate comparison across estimates, we normalize the standard deviation of all dependent variables to 1.

For the two physical boundaries, railroads and highways, all statistically significant boundary discontinuities are positive, which supports our selection procedure for H_{Cbj} (the “high side”). Moreover, the vast majority of economically significant discontinuities (defined as $\hat{\delta}_C > 0.05$) are statistically significant. The range of discontinuities for highways and railroads is similar, though railroad discontinuities tend to be slightly larger than highway discontinuities. For the two educa-

tional boundaries, we estimate even larger discontinuities than for the physical boundaries. Once again, all statistically significant boundary discontinuities are positive, and the vast majority of economically significant discontinuities are statistically significant. We estimate fewer statistically significant discontinuities at county boundaries, but a substantial fraction of statistically significant discontinuities at ZIP code boundaries. This is likely due to the fact that the majority of county borders are in rural areas that offer fewer observations for estimation, a drawback that does not apply to ZIP code boundaries.

Figure 2: Boundary Effects



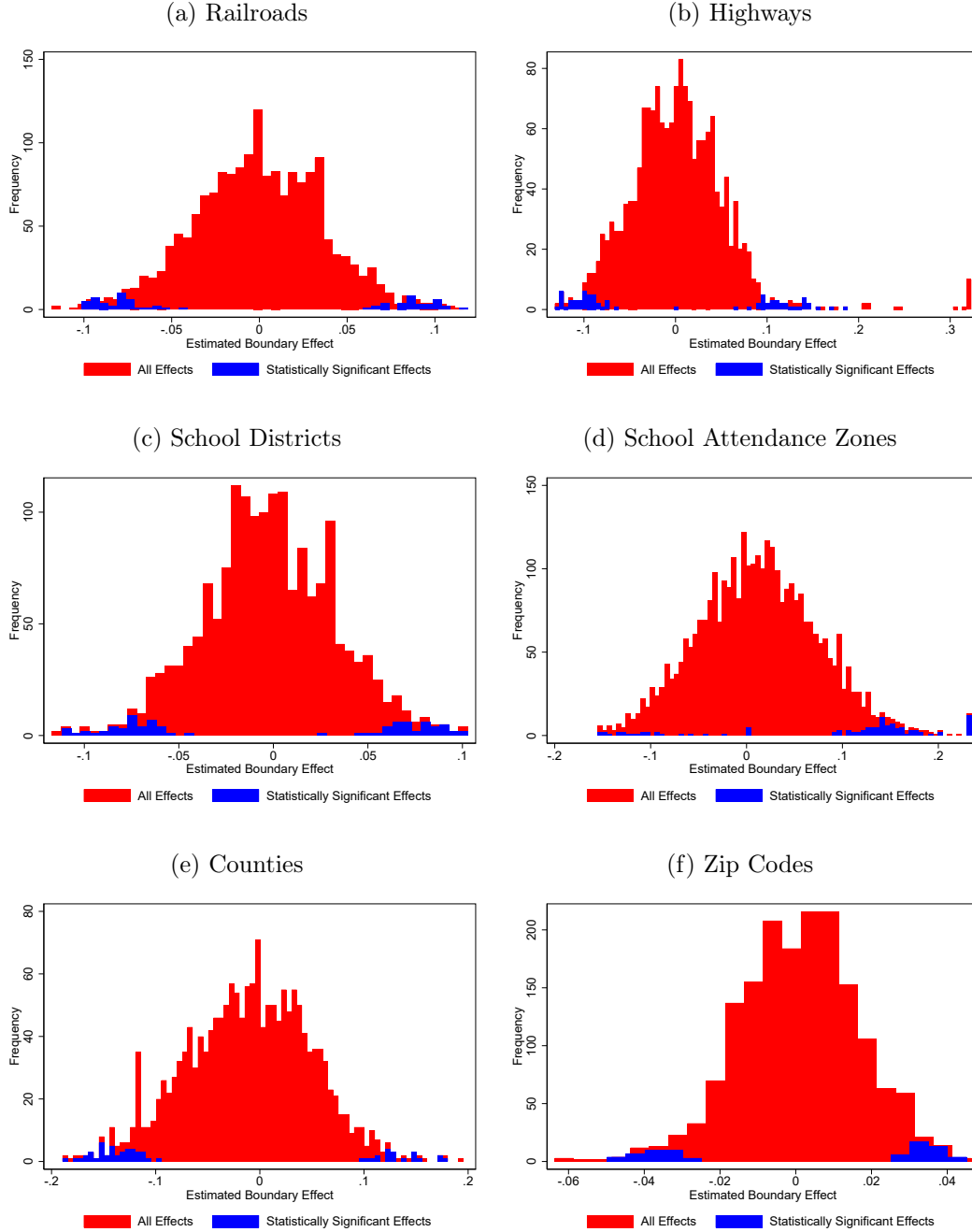
Note: Statistical significance is determined at the 95% level. Each histogram contains the frequency of statistically significant and insignificant effects arranged by the magnitude of the boundary effect (x-axis). For comparison across estimates, the standard deviation of all dependent variables is normalized to 1.

In order to ensure that our results are not simply a statistical artifact of our estimation and testing procedure, we reproduce these histograms using estimated discontinuities from our placebo validation exercise in Figure 3. Three observations are immediate for all boundary types: (1) The

overwhelming majority of placebo discontinuities are not statistically significantly different than zero. (2) The distributions of placebo discontinuities are roughly symmetric around zero.¹⁰ (3) The magnitudes of estimated placebo discontinuities are substantially smaller than the magnitudes of the estimated boundary discontinuities. All three observations strongly support our empirical approach.

¹⁰The slight skew to the right is likely due to our selection procedure for the “high side.”

Figure 3: Placebo Boundary Effects



Note: Statistical significance is determined at the 95% level. Each histogram contains the frequency of statistically significant and insignificant effects arranged by the magnitude of the boundary effect (x-axis). For comparison across estimates, the standard deviation of all dependent variables is normalized to 1. Detailed description of the placebo boundary exercise is in Section 3.

We summarize these results in the first column of Table 2. We report that 81% of characteristics change discontinuously at school attendance zone boundaries, followed by 76% at railroad tracks, 64% at ZIP code boundaries, and 61% at school district boundaries. Highways have less

Table 2: Fraction of Significant Effects

	All Vars.	Selected Vars.	With Controls	$\lambda =$ 0.55 miles	$\lambda =$ 1.05 miles
	(1)	(2)	(3)	(4)	(5)
Panel A: Fraction of Neighborhood Characteristics with Significant Boundary Discontinuities					
School Attendance Zone	0.810	0.841	0.715	0.724	0.779
Railroad	0.756	0.783	0.705	0.744	0.738
Zip Code	0.635	0.683	0.612	0.680	0.629
School District	0.610	0.643	0.513	0.687	0.675
Highway	0.508	0.531	0.366	0.575	0.556
County	0.167	0.178	0.112	0.223	0.234
Panel B: Average Size of Significant Boundary Discontinuities					
School Attendance Zone	0.403	0.406	0.263	0.303	0.326
Railroad	0.264	0.270	0.219	0.228	0.221
Zip Code	0.219	0.219	0.150	0.213	0.161
School District	0.262	0.304	0.213	0.262	0.233
Highway	0.307	0.311	0.230	0.233	0.209
County	0.471	0.499	0.468	0.161	0.212

Notes: We present statistics for $\hat{\delta}_C$ with various specifications and samples. Our default selection criterion of $\lambda = 2$ miles designates all neighborhoods as index neighborhoods for purposes of determining the high side (H_{cbj}) of each boundary segment. In column (2), we consider only the subsample of non-negative neighborhood characteristics that have fewer than 10% missing observations and no extreme outliers. In column (3), we add block group socioeconomic measures as control variables. Control variables include median income, the fraction of residents receiving public assistance, the fraction of owner-occupied housing, proxies for the median age and size of housing units, and the share of Black and Hispanic residents. In columns (4) and (5) we use alternate selection criteria of $\lambda = 0.55$ miles and $\lambda = 1.05$ miles to determine H_{cbj} . In column (6) we consider placebo boundaries. Each row summarizes the results of independent regressions taking each Census descriptive variable as an outcome, grouped by the particular boundary used in each set of regressions. Statistical significance is determined at the 95% level. All dependent variables have been normalized to have mean 0 and standard deviation 1.

of an influence than railroads, but affect over 50% of neighborhood characteristics. The weakest prevalence of boundary discontinuities occur at county lines. This heterogeneity is due to the spatial density of the different types of boundaries relative to the spatial density of neighborhoods.

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To ensure that these results are not driven by a large number of census variables being uninformative, in column (2) we consider only a subset of variables for which fewer than 10% of observations are missing, all entries are numeric, no observations accidentally take on negative

¹¹In Appendix A we present a more detailed description of the relative spatial densities of boundaries and Census Block Group centroids.

values, and no variables contain extreme outliers (a coefficient of variation over 2000). This eliminates roughly 20% of all neighborhood characteristics from our analysis. Discontinuities are found in a similar fraction of neighborhood characteristics.

In column (3), we re-estimate discontinuities controlling for socioeconomic measures that are strongly correlated with the quality of neighborhood amenities. Control variables include median income, the fraction of residents receiving public assistance, the fraction of owner-occupied housing, proxies for the median age and size of housing units, and the share of Black and Hispanic residents. We find that most observable neighborhood characteristics continue to vary discontinuously at neighborhood boundaries even conditional on these variables. This suggests that the use of observable controls is unlikely to truly account for omitted amenity bias affecting BDDs identifying assumptions.

In columns (4) and (5), we test the sensitivity of our results to our “high side” selection procedure. Specifically, we shrink the set of index neighborhoods that are used to determine the high side from all neighborhoods in a boundary segment (i.e., all neighborhoods whose centroids lie within 2 miles of a boundary) to all neighborhoods whose centroids lie within 0.55 miles and 1.05 miles of a boundary respectively. The fraction of observable neighborhood characteristics that exhibit boundary discontinuities changes very little under these alternative selection criteria. This lends support to our empirical approach.

Finally, in column (6), we re-estimate discontinuities from our baseline specification and sample (column 1) using placebo boundary locations. These results summarize the histograms shown in Figure 3. We find discontinuities in 3.9% to 5.3% of observable neighborhood characteristics at these placebo boundaries, which is what we should expect when considering significance at the 95% level.

In Panel B, we compare the average estimated effect size for statistically significant variables at each boundary type. For our baseline in column (1), the average magnitude ranges from 0.22 to 0.48 standard deviations of each observable characteristic. The larger discontinuities that we find for school zones reflect greater household sorting across school zones than other boundary types. Interestingly, county lines yield the fewest significant discontinuities but the largest average discon-

tinuity size. This may indicate that when a discontinuity exists at a county boundary, it tends to be large. Nevertheless, we find a similar prevalence and size of significant discontinuities as we did in the unrestricted sample of columns. Alternatives in columns (2)-(5) are broadly consistent. Finally, column (6) shows that even the few statistically significant placebo discontinuities that we do find are not economically significant. To further support our identification of widespread boundary discontinuities, we break down the distributions of estimates (including placebo estimates), and present sensitivity tests for RD bandwidth and polynomial choices in Appendix B.2.

4.2 Implications for Practice: When Boundaries Assign Treatment

4.2.1 Balance Tests

Our finding of robust, widespread boundary discontinuities has immediate implications for empirical designs that exploit geographic boundaries for identification. To start, researchers commonly assess pre-treatment data for continuity before estimating treatment effects. To support causality under the identifying assumptions of regression discontinuity designs, these balance tests provide evidence that neighborhoods immediately adjacent to the boundary are comparable except for the treatment itself (Keele and Titiunik, 2015). If pre-treatment balance is satisfied, post-treatment balance is strongest when potential confounders vary smoothly across the boundary space. Together the two balance tests provide strong support that a discontinuity in the outcome of interest can be attributed to the treatment itself.

While straightforward in theory, we speculate the power of balance tests is often limited by data availability. Researchers may possess necessary micro-data for an outcome (for example, repeat sales of a large number of individual houses near school attendance boundaries), but comparable data for the full set of neighborhood characteristics is unlikely to exist. Instead, balance tests thus rely on aggregate demographic information observed at a coarser spatial scale. Because boundary designs tend to focus on a single geographic region, a failure to detect discontinuities in confounding covariates may reflect statistical limitations rather than continuity in neighborhood characteristics.

To illustrate the potential for this limitation, we repeat our baseline analysis separately for 9 large Core-based statistical areas (CBSA) in the United States using school attendance bound-

aries. The selected CBSAs represent settings where school the school boundary data are relatively complete and therefore provide the most statistically powered subsamples. While the estimated effect sizes for each CBSA are similar to those of the nation as a whole, the precision of these estimates is dramatically smaller. Instead of finding statistically significant discontinuities in over 80% of neighborhood characteristics, we find statistically significant discontinuities between 10-40% of variables across CBSAs.

Table 3: School Attendance Zone Boundary Discontinuities by City

City (CBSA)	Observations	(1)	(2)
		Fraction of Statistically Significant Discontinuities (All)	Fraction of Statistically Significant Discontinuities, Selected Variables
All US (Pooled)	35604	0.810	0.841
Miami	2203	0.101	0.109
Philadelphia	1598	0.088	0.087
Minneapolis	1371	0.228	0.237
Houston	1344	0.223	0.234
Tampa	1141	0.184	0.188
Atlanta	1104	0.371	0.397
Riverside	1010	0.287	0.288
Washington, DC	958	0.163	0.163
Denver	855	0.118	0.102

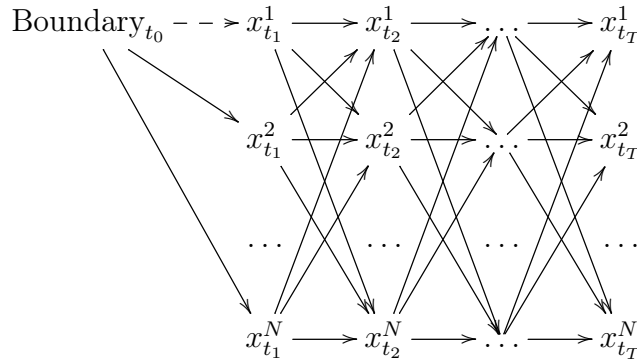
Notes: In column (1) we present statistics for $\hat{\delta}_C$ for the entire sample of neighborhood characteristics, and in column (2) we present statistics for $\hat{\delta}_C$ for the subsample of non-negative neighborhood characteristics that have fewer than 10% missing observations and no extreme outliers. Selected cities are chosen based on data coverage, i.e., the share of MSA population residing in a school zone available in GIS shapefiles. Sample cities are sorted in the table by total number of observations. Statistical significance is reported at the 95% level.

The results in Table 3 illustrate a practical limitation of balance tests in boundary designs. Researchers working with aggregate census data for a single city or metropolitan area lack the power to detect meaningful discontinuities in confounding variables. Therefore, covariate balance tests with aggregate data should be interpreted as suggestive, rather than definitive evidence that identifying assumptions hold. In the presence of convincing evidence of covariate balance, we next discuss how the sorting process described in Section 2 begins immediately after the introduction of a new boundary (4.2.2), or an exogenous treatment at an existing boundary (4.2.3).

4.2.2 The Canonical Design with New Boundaries

The gold standard application of boundary discontinuity design arises when a new boundary is introduced and both the treatment and the outcome are observed shortly afterwards. If pre-period balance tests are satisfied, implying the absence of overlapping boundaries and pre-existing breaks in geographical terrain, other neighborhood attributes remain continuous by construction. Thus, any subsequent discontinuity in the outcome can be attributed to the treatment itself.

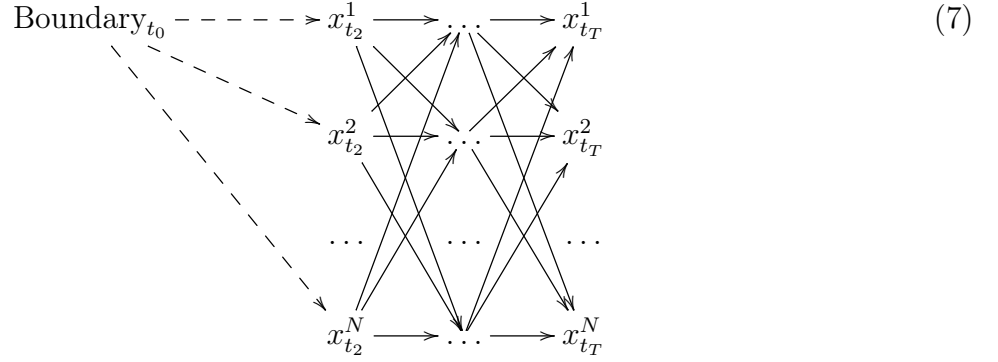
Consider the boundary is placed at period t_0 , with $\vec{x}$ representing a vector of neighborhood characteristics – both observed and potentially unobserved – in periods $t_1, t_2, \dots, t_T$.¹² Solid arrows indicate continuous causal relationships, and dotted arrows indicate discontinuous causal relationships. Suppose x^1 is a variable that the boundary is intended to directly influence (e.g., a local tax rate for an administrative boundary). Then we would expect a discontinuity in x^1 on either side of the boundary. While it is possible that the boundary might be correlated to other neighborhood characteristics, it need not generate a discontinuity in those characteristics. However, the sorting of people, firms and amenities in periods t_1 and thereafter imply dynamic linkages between the variables $x^1, \dots, x^N$. Nevertheless, a researcher could estimate the causal effect of an observed $x_{t_1}^1$ on any observed variable $x_{t_i}^n$, $n = 1 \dots N$, $i > 1$ using a boundary discontinuity design, as any discontinuity in $x_{t_i}^n$ at the boundary could only be due to $x_{t_1}^1$.



Suppose, however, that x^1 could only be observed starting in period t_2 . We would then re-draw

¹²In principle, t_0 could equal t_1 .

this diagram as



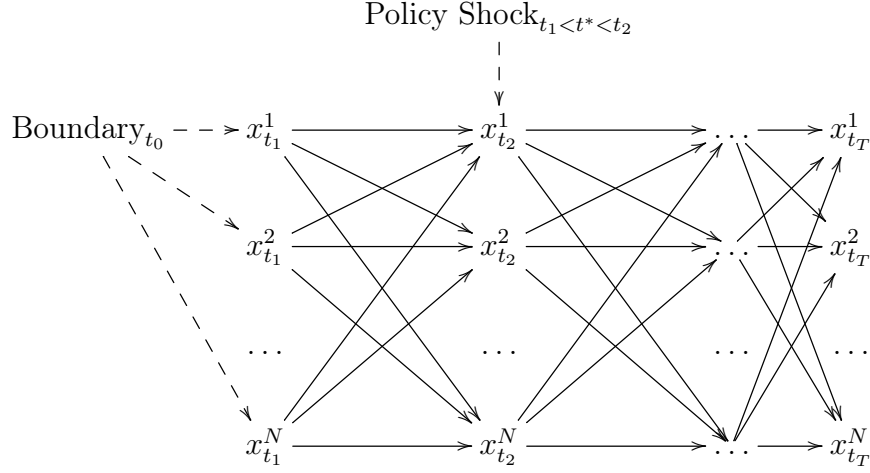
The identifying assumption weakens if households, firms, and local governments have time to respond before post-boundary data is observed. If so, the relationship between the boundary in t_0 and *all* variables in t_2 is discontinuous since these effects are mediated, at least in part, through the unobserved $x_{t_1}^1$. By t_2 the boundary no longer represents an exogenous influence only on x^1 , but a general equilibrium shift in the bundle of neighborhood characteristics that can evolve over time. This is the prevailing intuition behind studies examining contemporary outcomes at historic boundaries through the lens of institutions (Dell, 2010; Dell et al., 2018; Bühler, 2023). To assess persistence using boundaries requires understanding the sequence of mediating changes over decades of endogenous adjustments. Rather than an isolated effect of a single institutional attribute, the passage of time transforms identifying variation of a boundary from a single treatment into variation in an equilibrium neighborhood bundle.

4.2.3 New Policies at Existing Boundaries

A new policy affecting one side of an existing administrative boundary creates geographic thresholds frequently used to estimate local average treatment effects. Examples include literature on minimum wage policies at state or county level (Dubé et al., 2010; Jardim et al., 2024), place-based policies for economic growth (Jia et al., 2021a), and spillovers from public transportation investments (Li et al., 2016; Jia et al., 2021b). In these settings, the policy variation is amenable to spatial difference-in-differences and differences-in-discontinuities models (Butts, 2023).

Consider a policy directly targeting neighborhood characteristic x^1 . By the time implementation occurs at period t^* , the boundary has already generated Tiebout sorting dynamics among

households, firms and local governments. The policy creates a new discontinuity in x^1 , but does so in neighborhoods with equilibrium characteristics previously shaped by the boundary itself. Almost surely, a cross-sectional balance test using data from t_1 will fail due to existing boundary effects.



The identification problem is the boundary represents two roles simultaneously. At t^* the boundary determines treatment assignment for the new policy, while generating confounding variation through prior periods of endogenous sorting. As a result, exogeneity relies on the unanticipated timing of policy announcement as the boundary side receiving treatment is not plausibly random. Parallel trends help establish that treated and untreated neighborhoods follow similar trajectories prior to the policy, but do not imply the policy affects only the targeted margin of interest. Once the policy causes a trend break in x^1 , households and firms adjust, altering a broader set of neighborhood characteristics. The estimated treatment effect therefore captures both the direct effect of the policy and variation in the entire bundle of neighborhood amenities.

4.2.4 The Capitalization of Neighborhood Amenity Bundles

Hedonic models offer a way to evaluate equilibrium amenity differences at neighborhood boundaries. School attendance boundaries are particularly informative because they induce sorting along several dimensions (Bayer et al., 2007) and hence capitalize test score differentials (Black, 1999; Collins and Kaplan, 2022), student racial demographics (Wigger, 2025), and changes in the social-behavioral environments of schools (Crespin, 2024). As opposed to using house prices as a way

to assign monetary values to particular school (or neighborhood) attributes, we estimate the capitalized value of the full bundle of differentials observed in the main results. To illustrate this distinction, we estimate the unconditional house price premium for the high-amenity side of school attendance boundaries using nationally representative SABINS boundary data.¹³

Table 4 presents results from the hedonic analysis taking the log-transform of house prices as an outcome. Each column includes a sample restriction based on overlapping boundaries, and in column (5) we restrict the sample to segments with no overlap. In the upper panel of column (1) we estimate the a price differential of 26.4 percent when including all boundaries in the sample. The estimated premium reflects the equilibrium value households place on the combination of amenities coordinated by school attendance boundaries, including educational quality, student body and neighborhood composition, along with access to private consumption amenities. Moving from left to right in the table we address overlapping boundaries, a sensitivity check due to the fact that almost 60 percent of block groups from column one are influenced by other boundary types. When school attendance boundaries stand alone, as shown in column (5), the estimated premium rises to 32.9 percent.

Table 4: Hedonic Estimates at School Zone Boundaries

	All	No District Effect	No RR Effect	No HWY Effect	No Effects 2-4	With Boundary FE
Log(Price)	(1)	(2)	(3)	(4)	(5)	(6)
RD Estimate	0.264**	0.251*	0.256*	0.329***	0.329*	0.105**
SE	(0.0894)	(0.0993)	(0.106)	(0.0983)	(0.128)	(0.0350)
<i>N</i>	14223	9954	9395	10558	5704	5704

Notes: Each column represents estimates of the unconditional house price premium at school attendance zones. Each regression models distance to the boundary as a third-order polynomial following the Imbens and Kalyanaraman (2012) optimal bandwidth selection procedure. Statistical significance is reported at the 95% level.

Given the large, geographically dispersed sample of boundaries in the data, these estimates reflect the national average price premium in the midst of a complex, dynamic, US public school system with many actors and agents facing a variety of changing constraints. Naturally, the result is markedly larger than estimates for the capitalization of test-scores alone. Employing boundary fixed effects, Black (1999) presents hedonic estimates of 2.1 percent for a 5 percent

¹³In this exercise, the high-side selection procedure simply involves choosing the side of each boundary with the higher average home values in the 2010 Census.

increase in campus test scores, and Bayer et al. (2007) estimates 1.8 percent conditional school quality premium. Such fixed effects are useful to absorb the amenity differences we highlight in an effort to obtain the value of school quality alone. Even with the inclusion of fixed effects, we report a 10.5 percent full-on price premium in column 6.¹⁴

5 Conclusion

Physical structures, both natural and man-made, distort the urban landscape. So too do administrative boundaries that allow for the differentiation of public goods. In this paper, we present a simple model that yields the prediction that these distortions will manifest as discontinuities in neighborhood characteristics across boundaries of many types. We then show that a comprehensive set of neighborhood characteristics – the universe of publicly available characteristics in the decennial Census – exhibit discontinuities at a broad set of physical, educational and administrative boundaries. These discontinuities are sizable, systematic, and not merely statistical artifacts of how spatial data are collected.

Given these findings, we argue that the popular boundary discontinuity design should be applied with caution as its core identifying assumption may not hold in certain settings, and a standard validation exercise of this assumption is probably under-powered to draw a meaningful conclusion in practice. This yields an insight that should be taken to heart by both policymakers and researchers. Although the short-run effects of boundaries may be narrow, the long-run effects of boundaries are likely to be broad in scope, even if the treatment induced by the boundary is very narrow. Shifting a school attendance boundary has the potential to affect far more than educational outcomes; adding a highway will affect neighborhoods in far more profound ways than changing traffic patterns; past institutional boundaries such as redlines that are no longer in effect may still generate dramatic discontinuities in the present day. Policymakers would be wise to consider these knock-on effects when assessing if and where to place boundaries. And researcher should perhaps trade-off the

¹⁴Black (1999) finds that a 5 percent increase in test scores changes house prices by 4.9 percent for the full sample and by only 2.1 percent when controlling for boundary fixed effects. Bayer et al. (2007) similarly finds boundary fixed effects reduce the capitalization of school quality by 50 percent, and states that boundary fixed effects in a hedonic price regression are not fully effective in controlling for all aspects of neighborhood quality.

hope of using boundaries for the sharp identification of narrow treatment effects for the prospect of using boundaries to explain a broader set of spatial phenomena.

As an illustrative example we unpack the price house price capitalization of Tiebout sorting at school attendance boundaries, and find discrete price changes close to five times the magnitude of estimates for the capitalization of school quality alone. The large price differential reflects broad changes in the bundle of amenities, including school quality and social networks that influence long-run labor market outcomes. While school quality equalization is worthwhile to pursue, the sorting we find along a number of different margins (at various boundary types) suggests that such a policy addresses only a portion of location choices that influence lifetime outcomes.

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Online Appendix

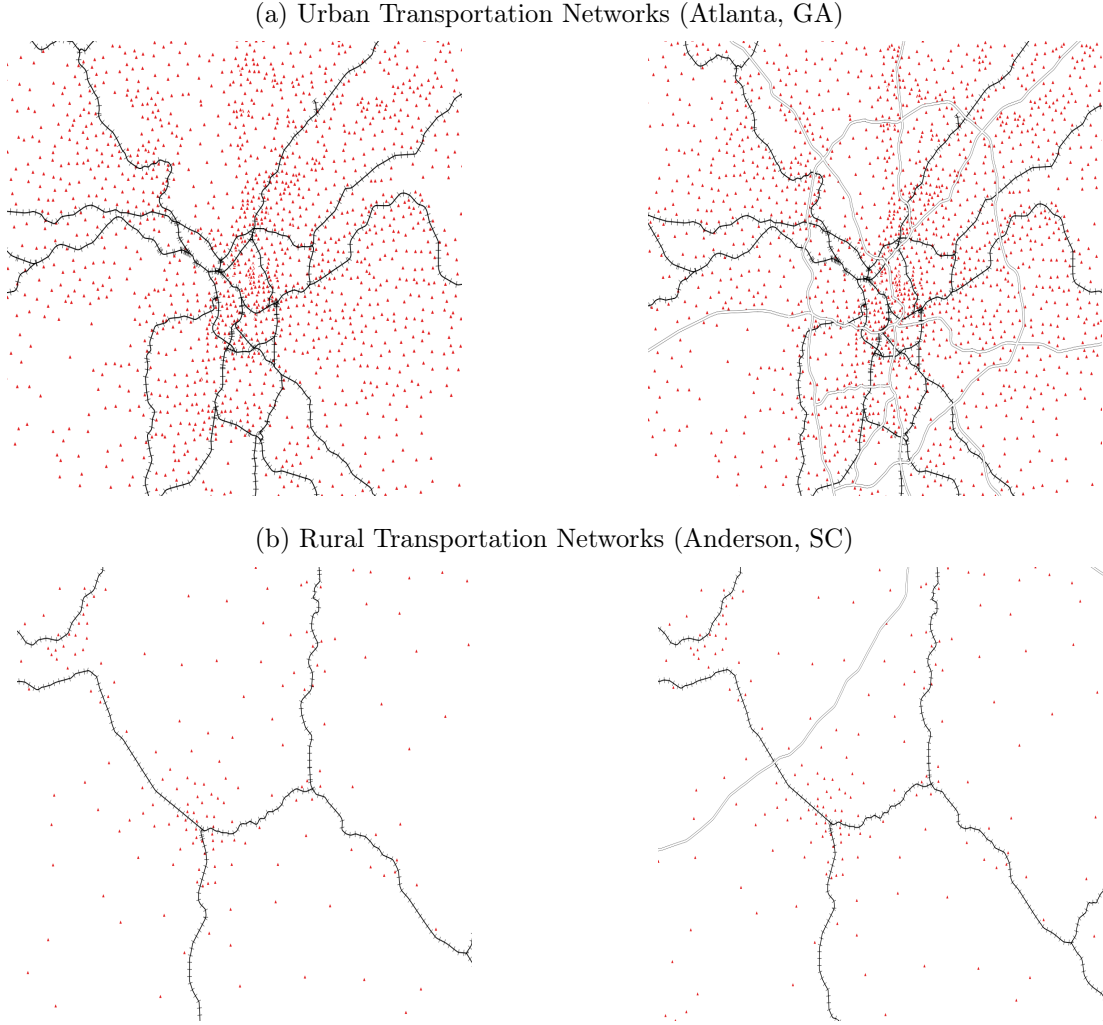
A Data and Sampling

The approach to constructing our data is the same for each of the six boundaries. Original source data shapefiles are loaded into ArcGIS, and each boundary is cut into segments at maximum two miles in length. Block groups are assigned to nearby boundary segments using the latitude and longitude coordinates of each Block group population center. For each boundary type, we allow Block groups to be assigned to multiple boundary segments. Although Block groups are discrete points across space, the data proxy for a continuous distribution of amenities near each boundary. In that sense, a single Block group that is 0.75 miles from one boundary and 0.1 miles from another can provide valuable information about each. The sample is restricted to Block groups less than a mile from a boundary, and boundary segments must have data on both sides to be included.

A.1 Physical Boundaries

The data for railroad and highway networks is comprehensive and covers the entire United States. The sum total of railroad mileage is greater than highway mileage, and the arrangement of both networks may differ in urban and rural areas. As an example, Figure 4 contains a comparison of railroad and highway coverage in a large urban area (Atlanta, GA) and a suburban/rural county (Anderson, SC) in the same region. Evident from the upper panel are transportation networks serving as major arteries in the city structure, with substantial clustering of neighborhoods near both railroads and highways. Economic activity in rural areas is more likely to be clustered near railroads, as shown in the the lower panel of Figure 4.

Figure 4: Railroads and Highways



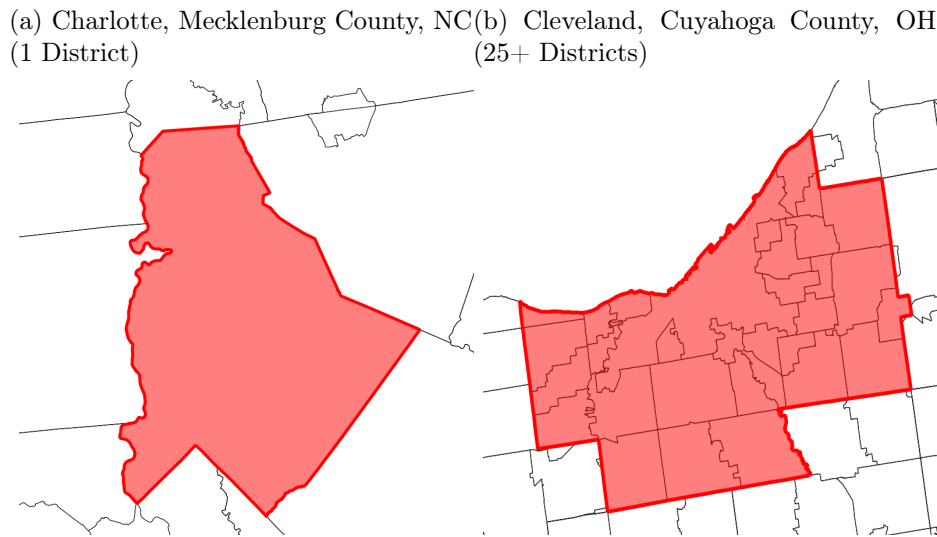
Notes: In each panel, the left Figure includes only railroad lines, while the right Figure includes both railroad and highway lines. Railroad lines are colored solid black and highway lines are outlined in grey.

A.2 School Boundaries

There is an extensive body of research on the formation and consequences of school boundaries. There is substantial heterogeneity in the political economy behind how school district and attendance zone boundaries are drawn by state. For the two school boundary types, we try to use the broadest definition possible. The school district sample includes unified districts that do not solely serve charter schools. School attendance zones in the sample include those that contain third grade students and are intended to have open enrollment. Open enrollment school zones allow students with the catchment area to choose from two or more schools that share the same boundary.

School district maps can vary substantially by state, and are a classic example of variation in the political economy that has historically shaped school districts. On one extreme are school districts drawn synonymously with county lines, covering large spatial areas and governed by a single school board. Such design is prominent in southern states. By contrast, other counties are fragmented with smaller school districts each managed by an independent school board. In the latter case school district lines may coincide with municipal boundaries at a lower geography than the county. As an example, Figure 5 contains two counties (shaded areas) with similar populations in 2020. On the left is Mecklenburg County, population 1.13 million, in which Charlotte is center city. On the right is Cuyahoga County, population 1.23 million, where Cleveland is the center city. As of 2020, Cuyahoga county is served by over 25 school districts.

Figure 5: School District Heterogeneity



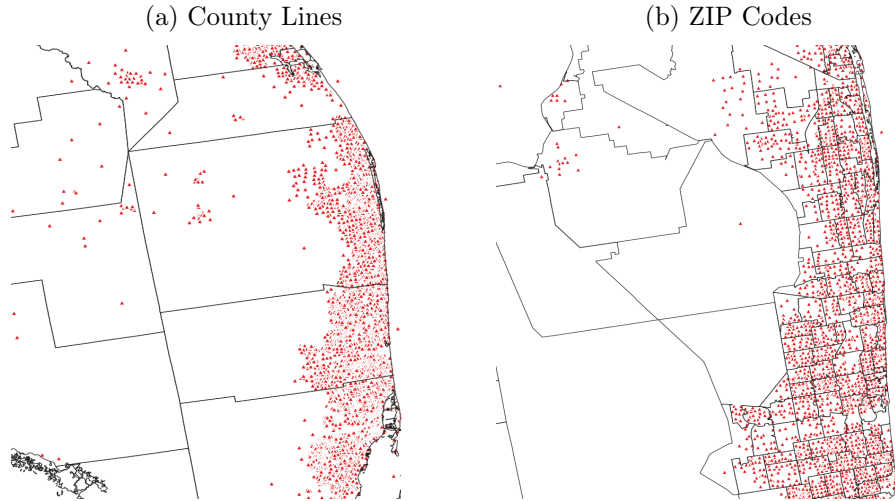
Notes: Mecklenburg County, NC (4.1) has a population of 1.13 million in 2020, in comparison to the 1.23 million residents of Cuyahoga County, OH (4.2). The upper panel illustrates the single unified Charlotte-Mecklenburg school district in contrast to the many districts serving the Cleveland-Cuyahoga County area.

A.3 Administrative Boundaries

Data for county lines and ZIP codes cover nearly entire lower 48 geographic area. As Figure 6 shows, counties have broader administrative geography with boundary sections in both urban and remote areas. There are over 3,000 US counties, and across county lines are predicted changes in tax rates levied on property and sales, municipal service quality, and school quality. There is substantial

heterogeneity in county population sizes, with the average US county being approximately 100,000 people.

Figure 6: Example: Greater Miami MSA



Notes: Census Block group locations are overlaid with spatial data for county lines and ZIP code boundaries in the Greater Miami metropolitan area, including Dade, Broward, and Palm Beach counties.

As of the 2020 census there are 41,500 zip codes containing an average 8,340 people in each. As shown in the second panel of Figure 6, the spatial correlation between ZIP code density and population density combined with a contiguous ZIP code map yields a substantially larger analysis sample of ZIP code boundaries. The primary purpose of ZIP codes is to delineate federal postal routes, in theory an otherwise meaningless distinction that in practice is used to facilitate other locally provided services. Public school catchments, police and fire precincts, county lines and other municipal boundaries have the potential to overlap with ZIP codes to produce statistically significant amenity differentials in our analysis.

B Supplementary Analysis

B.1 Monte Carlo

For each of 5000 boundary segments indexed by j , we sample 100 neighborhoods indexed by i that are characterized by a location $x_{ij} \sim U(-10, 10)$ that represents the distance of the neighborhood from a boundary, and a characteristic shock $\epsilon_{ij} \sim N(0, 0.5)$. The characteristic of each neighborhood is given by:

$$c_{ij} = \beta x_{ij} + \delta \times 1_{x_{ij} > 0} \times f_j + \epsilon_{ij} \quad (8)$$

where $f_j = 1$ with probability $\frac{1}{2}$ and -1 with probability $\frac{1}{2}$. Intuitively, there is a general spatial trend in neighborhood characteristics that is parametrized by β , there is a boundary discontinuity parametrized by δ , and the parameter f_j randomly “flips” the orientation of the data in each boundary segment, i.e., the “high side” of the data is randomly determined.

We simulate these boundary segments for a variety of chosen values of β and δ . In each simulation, we empirically determine the high side of each boundary segment, $\tilde{f}_j$, by comparing the mean of y_{ij} for all neighborhoods within 1 unit of the boundary on either side. That is, $\tilde{f}_j = 1$ if the mean of y_{ij} for all neighborhoods with $0 < x_{ij} < 1$ exceeds the mean of y_{ij} for all neighborhoods with $-1 < x_{ij} < 0$. We then estimate $\hat{\beta}$ and $\hat{\delta}$ by least squares. Estimates of $\hat{\delta}$ for a variety of simulated parameter values are presented in table XXX. Estimates of $\hat{\delta}$ for $\delta = 0$ are insignificant at the 95% level; all other estimates of $\hat{\delta}$ are significant at the 95% level.

TABLE

We also tried various other nonlinear specifications of $c_{ij}(x_{ij})$ (quadratic, absolute value, log, absolute value of the log) and obtained similar results.

B.2 Sensitivity Analysis

B.2.1 Polynomial Choice

As standard with regression discontinuity designs we stress test our results for the choice of RD polynomial. Table 5 presents results modeling each outcome as a linear function of distance to the

boundary (as opposed to the cubic polynomial functional form in the main results). In comparison to Table 2, the results are qualitatively similar in the ranking of boundary types, however, the linear specification overstates the prevalence of Tiebout sorting effects.

Table 5: Summary of Results

	Frac. Significant Effects		Avg. Significant Effect Size	
	All Vars.	Selected Vars.	All Vars.	Selected Vars.
	(1)	(2)	(3)	(4)
Panel A: Actual Boundaries				
School Attendance Zone	0.902	0.948	0.339	0.359
Railroad	0.829	0.871	0.194	0.209
Zip Code	0.752	0.793	0.144	0.151
School District	0.721	0.754	0.191	0.203
Highway	0.609	0.656	0.158	0.165
County	0.278	0.289	0.135	0.148

Notes: In columns (1) and (3), we present statistics for $\hat{\delta}_C$ for the entire sample of neighborhood characteristics. In columns (2) and (4), we present statistics for $\hat{\delta}_C$ for the subsample of non-negative neighborhood characteristics that have fewer than 10% missing observations and no extreme outliers. Each row summarizes the results of independent regressions taking each Census descriptive variable as an outcome, grouped by the particular boundary used in each set of regressions. Statistical significance is reported at the 95% level.

B.2.2 Distance Bandwidth

As in traditional RD settings, the primary hurdle in spatial RD design is determination of the bandwidth, defined by the distance from the boundary on either side that determines which data points to be included in the estimation sample. A common practice is to repeatedly estimate the sharp RD model, reducing the sample each iteration by shrinking the bandwidth. We follow Imbens and Kalyanaraman (2012) bandwidth selection process that minimizes an empirical approximation of the mean squared error. Using the Calonico et al. (2017) RDRobust package implements Imbens and Kalyanaraman (2012) optimal bandwidth selection for each regression outcome in the main analysis.

Table 6: Summary of Results

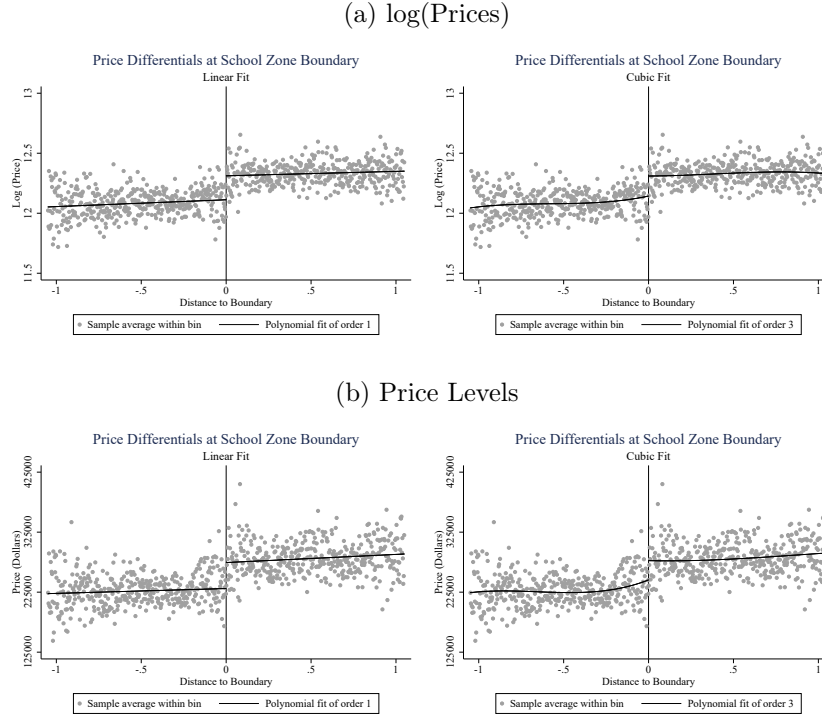
	Frac. Significant Effects		Avg. Significant Effect Size	
	All Vars.	Selected Vars.	All Vars.	Selected Vars.
	(1)	(2)	(3)	(4)
Panel A: 1 Mile Bandwidth				
School Attendance Zone	0.896	0.942	0.325	0.343
Railroad	0.812	0.849	0.186	0.194
Zip Code	0.744	0.781	0.140	0.148
School District	0.712	0.745	0.184	0.191
Highway	0.606	0.634	0.136	0.311
County	0.191	0.196	0.114	0.119
Panel B: 0.5 Mile Bandwidth				
School Attendance Zone	0.040	0.036	0.086	0.075
Railroad	0.046	0.045	-0.012	-0.007
Zip Code	0.048	0.048	-0.001	0.006
School District	0.050	0.050	-0.045	-0.019
Highway	0.053	0.047	-0.037	-0.003
County	0.039	0.043	-0.027	-0.025

Notes: In columns (1) and (3), we present statistics for $\hat{\delta}_C$ for the entire sample of neighborhood characteristics. In columns (2) and (4), we present statistics for $\hat{\delta}_C$ for the subsample of non-negative neighborhood characteristics that have fewer than 10% missing observations and no extreme outliers. Each row summarizes the results of independent regressions taking each Census descriptive variable as an outcome, grouped by the particular boundary used in each set of regressions. Statistical significance is reported at the 95% level.

B.3 Supplement : Hedonic Regressions and Boundary Fixed Effects

Figure 7 includes raw data scatter plots of census block group home prices for owner-occupied housing. Prices are displayed in logs on the y-axis, and the x-axis arranges block groups by distance to the boundary. For each boundary segment, the median house price is computed for each side, and block groups on the higher income side take positive values. Each panel includes identical underlying data but vary by the polynomial degree chosen for the line of best fit. In a sharp RD design, the point estimate is the difference between two regression functions evaluated at the boundary, represented by the vertical difference the two fit lines. Comparing the panels illustrates how the potential for misspecification if non-linearities in the house price gradient go unmodeled.

Figure 7: Mean Home Values by Distance to the Boundary, 2010 Census



Notes: Scatterplots of raw data for Census Block groups arranged by distance to a school attendance zone boundary. Each Figure varies only by the polynomial selected to model the local regression on each side of the boundary. Point estimates in a hedonic model of prices are computed as the vertical distance between the two regression lines as they intersect the boundary.

Non-parametric estimation of house price changes at school boundaries captures the general equilibrium effect of better quality schools, neighborhood demographics that change with school quality, and other amenities correlated with neighborhood demographics. This contrasts with using school boundaries and house prices to estimate household valuation of school quality while conditioning on demographic and amenity differences. On one hand, it is important to know the value that households place on educational investments. On the other hand, the overall general equilibrium effect is informative as to the amount households must pay to access better quality schools. In places without open enrollment, i.e. school attendance zones are binding, households must be willing to pay for both higher school quality and the bundle of neighborhood demographics and amenities that accompany better schools.

Better educated families with higher incomes may sort into high quality school zones for a variety of reasons. Thus, the demand for housing near high quality schools also reflects the fact

that better educated neighbors with higher incomes are positive neighborhood amenities, and other amenities near such neighbors are also likely to be of higher quality. Households choosing where to live may find these neighborhoods desirable over and above the quality of their schools. Our results in the main body of the paper suggest that households must be willing to pay for both the higher school quality and the price capitalization of other amenities.

Boundary fixed effects are a way to de-mean house prices in block groups relative to neighbors in the same boundary segment. In a hedonic model this approach compares residual variation in prices to residual variation in an observable neighborhood characteristic to obtain a hedonic valuation of that particular amenity. Incorporating boundary fixed effects into a local linear RD involves two stages : regressing house prices on a set of indicators for each boundary segment, then fitting the RD model with residuals from the first stage. In the main results we estimate the hedonic RD taking the natural log of census home values as an outcome, comparing results with and without boundary fixed effects. A similar exercise for the price level is carried out in Table 7.

Table 7: Hedonic Estimates at School Zone Boundaries

Outcome: Price Level	All	No District Effect	No RR Effect	No HWY Effect	No Effects 2-4
Sharp RD	(1)	(2)	(3)	(4)	(5)
Estimate	53211.4*	48953.9	57567.3*	68503.9**	69223.3*
SE	(23059.2)	(26184.0)	(27918.4)	(25421.7)	(33840.9)
Boundary Fixed Effects RD	(6)	(7)	(8)	(9)	(10)
Estimate	27309.2***	25429.3***	34503.7***	29258.2***	34296.9***
SE	(6835.9)	(7664.5)	(8585.0)	(7595.4)	(9863.1)
<i>N</i>	14223	9954	9395	10558	5704
Polynomial	3	3	3	3	3

Column 1 of Table 7 includes all school zone boundaries, and in the subsequent columns we omit school zone boundaries that overlap with other boundary types. The results in the top panel highlight the effect of overlapping boundaries on the estimate of interest. This source of omitted variable bias is absorbed by boundary fixed effects, and in the bottom panel we show that the inclusion of fixed effects yields more consistent results. However, the tradeoff is elimination of interesting variation in the gross price differential, or the full on board cost of living in a higher quality school zone.